



Refocusing multiple stressor research around the targets and scales of ecological impacts

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Ecological communities face a variety of environmental and anthropogenic stressors acting simultaneously. Stressor impacts can combine additively or can interact, causing synergistic or antagonistic effects. Our knowledge of when and how interactions arise is limited, as most models and experiments only consider the effect of a small number of non-interacting stressors at one or few scales of ecological organization. This is concerning because it could lead to significant underestimations or overestimations of threats to biodiversity. Furthermore, stressors have been largely classified by their source rather than by the mechanisms and ecological scales at which they act (the target). Here, we argue, first, that a more nuanced classification of stressors by target and ecological scale can generate valuable new insights and hypotheses about stressor interactions. Second, that the predictability of multiple stressor effects, and consistent patterns in their impacts, can be evaluated by examining the distribution of stressor effects across targets and ecological scales. Third, that a variety of existing mechanistic and statistical modelling tools can play an important role in our framework and advance multiple stressor research.

Habitat loss and degradation, pollution, temperature changes and harvesting are some of the biggest stressors affecting ecosystems around the world¹. This list of stressors has become standard throughout ecology and conservation, from undergraduate teaching to international policy documents, and broadly groups ecosystem threats by where stressors come from—by their source. This is explicitly stated by current programmes for classifying the effects of different stressors, such as the Living Planet Index (LPI) and the International Union for Conservation of Nature (IUCN) Threats Classification Scheme, which define threats as “synonymous with sources of stress and proximate pressures”^{2,3}. Such ‘source-based’ classifications of stressors are certainly necessary for understanding the economic space in which mitigation can be applied and policy developed. They are effective, for example, at offering descriptions of how frequently taxa are being affected by a source of stress, which in turn can help make it clear where management actions should be targeted. However, they provide little insight into the mechanisms and ecological scales through which stressors act, which in turn hinders our ability to identify commonalities in how stressors re-shape ecological communities. Thus, source-based schemes alone, while valuable, may be insufficient to manage and mitigate threats to biodiversity and ecosystem function. Consider the example of pollution and temperature change. These stressors are generally regarded as separate under a source-based classification. However, when considering the mechanisms by which they operate, pollution and temperature might be grouped together as stressors that act primarily by altering the metabolism and physiology of individuals. Such a grouping makes it easier to understand

how and why these stressors might interact, leading to more accurate estimates of the prevalence, magnitude and direction of their combined effects.

Improving our understanding of potential interactions between stressors is critical because stressors often co-occur in time and space, resulting in more-than or less-than additive effects (synergistic or antagonistic interactions, respectively)^{2–10}. These interactions can occur as interaction chains, whereby one stressor increases the level of a second stressor, but the per capita impact of the second stressor does not change (such as habitat loss increasing introduced species abundance). Alternatively, they can occur as interaction modifications, whereby the per capita impact of a stressor changes with the level of a second stressor (such as habitat loss changing the per capita interaction strength between native and introduced species)¹¹. Without quantification of stressor interactions, we risk significantly overestimating or underestimating threats to biodiversity. Despite their importance, our current understanding of stressor interactions is limited, with studies generally only considering one or two stressors at a time, often ignoring interactive effects and considering only one scale of ecological organization^{12–15}. Measuring the magnitude and distribution of multiple stressor effects across ecological scales therefore remains a persistent challenge and a necessary objective. Ideally, we need a way to understand how accurately impacts of many different stressors can be forecasted (predictability) and the extent to which there are consistent patterns in stressor impacts in and among ecological scales (consistency).

Here, we argue for classifying stressors by the ecological scales at which they have their impacts rather than by their source. We use

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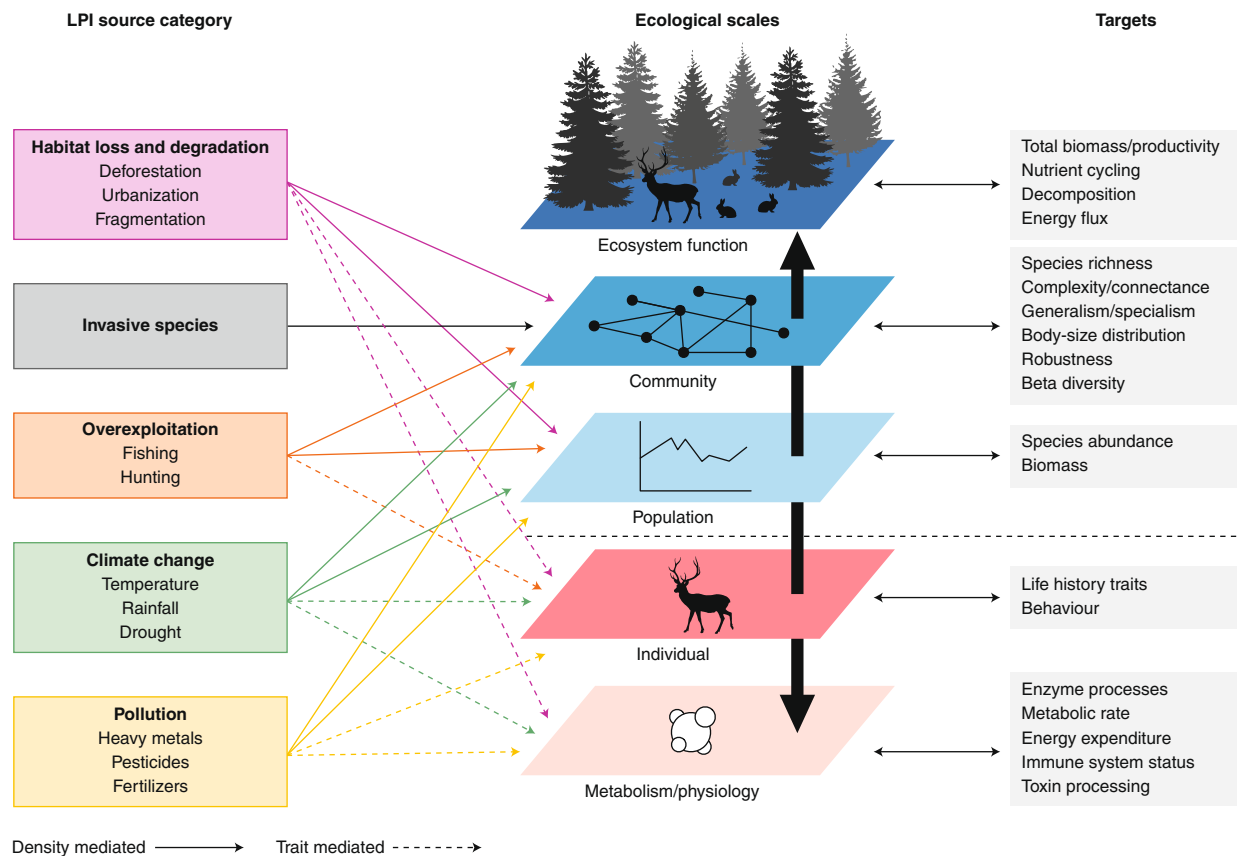


Fig. 1 | Conceptual diagram of the ecological scale-target-based classification of stressors to quantify the impact of multiple simultaneous stressors on ecological communities. We align the LPI source-based categories of stressors with a classical view of ecological scales and the target metrics at each scale. The framework (and models, see below) highlights targets and ecological scales that coalesce the impacts of stressors by what they do, not what they are. For example, pollution and climate stressors can both generate direct effects on the abundance of species by killing individuals, but also via trait-mediated effects acting on growth and reproduction. As different ecological scales are connected by higher-order interactions, outputs at higher and lower levels of ecological organization can emerge. Note that we define the effects of a stressor at the individual scale as those that act within, rather than between, individuals. Thus, a harvesting process, such as hunting, can have individual-scale effects by altering stress hormone levels in individuals, which has consequences for demography and therefore the population. However, hunting primarily acts by killing individual animals or plants, which directly affects properties of the population (abundance/biomass) not properties of the individual. Note that this figure is a first attempt designed to illustrate our framework and is by no means comprehensive.

this mechanistic classification of stressors to advocate focusing on the distribution of additive versus interactive stressor effects across different scales of ecological organization (from the physiology of individuals to populations, communities and whole ecosystems). We argue that thinking about this distribution is a common ground for theorists and empiricists alike and can be used to quantify the predictability and consistency of multiple stressor effects. Given the difficulty of large-scale experimentation with multiple stressors, we outline a set of criteria we believe that the next generation of ecological models must satisfy to generate informative predictions for improving knowledge of the impacts of multiple, interacting stressors. Finally, we review several approaches to building such model frameworks alongside statistical approaches for analysing their outputs and related empirical data.

The ecological scales of environmental change

In nature, there are many different scales at which stressors generate impacts: metabolism and physiology, individuals, populations, communities and ecosystems (Fig. 1). At each of these scales, there are a number of targets in which we can detect the effects of stressors. Encompassed by a wide range of ecological theory, these targets include processes regulated by enzymes and metabolic rates at the physiological scale, the life history traits of individuals, the

abundance or biomass of a population, or the diversity or structure of a community (Fig. 1). Targets are therefore both the ‘access points’ through which stressors enter ecological systems and the response variables used to quantify the additive or interactive impacts of stressors.

This idea of defining properties of organisms and scales as the targets of a stressor is most well developed in ecotoxicology by the ‘mode of action’ concept^{16,17} and the source–pressure–pathway–receptor models in cumulative risk assessment¹⁸. To a lesser extent, the idea has also been discussed in ecology, with Schäfer and Piggot recognizing three stressor modes of action¹⁹, Orr et al. suggesting that the similarity of stressor modes of action could generate insight into stressor interactions⁷, and Boyd and Brown recognizing the different scales at which stressor interactions can occur²⁰. Other works have also laid an important foundation for the mode of action approach^{21–23}, although these tend to be system-specific or scale-specific.

There have also been modelling approaches that have provided frameworks to integrate stressors on the scales of individuals, populations or communities. Such approaches often span ecological scales (for example, individual, population or ecosystem function), but they typically do not represent diverse networks of interacting species and they predict and evaluate only a limited set of end

points (targets). As such, they do not provide the opportunity to assess predictability and consistency. For example, de Laender et al.²⁴ provide a model centred on a single trait (resource uptake), focusing on competitive interactions and derived from core theory in plant ecology and productivity-diversity and diversity-stability research. While the study uses a multi-species and multi-stressor model, it is essentially focused on a single trophic level and a narrow set of traits. Goussen et al.²⁵ offer another approach that combines dynamic energy budget theory with an individual-based model. The study allows several traits to be defined in the dynamic energy budget to respond to stressors, but focuses on single species. Finally, Liess et al.²⁶ propose a model focused on a single population level trait (mortality) and propose an additive null model of stressor effects on a concentration-response relationship. While this is a useful approach to dealing with toxicants, it is not generalized across multiple traits, stressors and scales.

These previous works are highly valuable and lay the foundation from which we argue that only by focusing on targets can we move towards reliably estimating the magnitude and direction of multiple stressor effects at multiple ecological scales. We therefore argue for a multi-trait, multi-species, multi-scale approach to detecting interactions and evaluating predictability and consistency. By grounding this in a food-web framework, we also intrinsically incorporate multiple trophic levels in a community (rather than just one). Such an approach is distinguished from the taxa-, scale- and system-specific attempts described above by facilitating generalization and an evaluation of consistency across systems, stressors and scales.

Visualizing the framework. The left-most column of Fig. 1 shows a traditional source-based classification of stressors from the LPI, whereby habitat loss and degradation, invasive species, overexploitation, climate change and pollution are treated as separate stressors. Assigning these sources of stress to the ecological scales and targets at which they act reveals several patterns central to understanding the predictability and consistency of multiple simultaneous threats (Fig. 1).

The first pattern to recognize in Fig. 1 is that because most sources of stress directly impact on many ecological scales²⁷, the scales and their targets reconcile how disparately defined stressors act via similar mechanisms and on similar targets. This single change in point of view reduces the emphasis on the identity of the stressor and accentuates the long-standing ecological principles by which they have an impact. This means that seemingly separate stressors entering at the same scale can be evaluated by the same body of physiological or ecological theory because the stressors are defined by their targeting of similar properties, traits or biological mechanisms rather than by their source.

The following bodies of theory are relevant for our framework. Energy allocation and dynamic energy budgets^{13,28}, stress physiology and ecotoxicology¹⁶, and temperature scaling²⁹ all underpin our understanding of how sublethal effects of contaminants and temperature alter the abundance of species. The functional response³⁰ underpins our understanding of how harvesting (for example, fishing and hunting) impacts directly on the biomass and abundance of species. The concepts of trait-mediated versus density-mediated effects^{31,32} link these two collections of theory. Species interactions, trophic indirect effects, trophic cascades and stability^{32–34} all underpin our assessment of how perturbations to the numbers of individuals in populations permeate entire communities. Productivity-diversity/stability theory^{35–38} underpins our understanding of how enrichment of producers or whole trophic levels impacts on community stability and diversity. Disturbance theory, beta-diversity and species distribution theory speak to the temporal scale that stressors operate on (discrete versus continuous stressors events)³⁹. Finally, species distributions and diversity theory^{40–43} facilitate inference about processes operating on spatial scales.

This breadth of theory is fundamental to framing inference about the magnitude, direction and distribution of stressor interactions, as the theories define the traits and rationale for when and why interactions arise. As a specific example, consider temperature and sublethal concentrations of pesticides. Both stressors ultimately target metabolic processes that mediate the allocation of resources to growth and reproduction. At a lower mechanistic level, temperature is affecting all enzymatic processes while toxins such as pesticides are often categorized by reproductive, digestive or neurological modes of action. However, recognizing a common ecological scale and a common currency (energy allocation¹³) reveals focused bodies of theory and data on which to frame inference about when and how interactions might arise, their magnitude and at what scale they may arise^{24–26,44,45}. In this case, we might draw on the rich history of resource allocation theory, dynamic energy budget theory and trait-mediated direct and indirect effects theory^{31,32,46,47} to generate insights that are simply not possible when thinking about stressors in terms of their source alone.

The second pattern to recognize in Fig. 1 is that most sources of stress act via two pathways: (1) they generate density-mediated effects, that is, via directly affecting density and abundance, and (2) trait-mediated effects, that is, 'sublethal' effects generating change in the abundance and distribution of organisms via alterations of life history, behavioural traits and traits that define metabolism or physiology. In our framework, trait-mediated effects encompass changes at the individual scale, such as life history and behaviour, and changes at the metabolism/physiology scale, such as enzyme processes, metabolic rate, energy expenditure, immune system status and toxin processing. Thus, for example, oxidate stress is treated as a trait-mediated stressor that affects physiological targets. This link to classical ecological theory synthesizes the diversity of ways stressors can exert impacts alone and interactively. For example, drought can have density-mediated effects via mortality on single populations or multiple populations (the community), or act via trait-mediated effects at the individual or physiological scale, thereby reflecting other outcomes such as impaired reproduction, dispersal or diapause. This also highlights that the spatial and temporal scale that a stressor acts on can affect the ecological scale and pathway where the impacts of stressors arise⁴⁸. For example, a discrete extreme heat event might have density-mediated effects at the population and community scales, while continuous ongoing warming temperatures might initially have trait-mediated impacts acting at the metabolism/physiological scale.

Finally, the third pattern to recognize is that the effects of stressors can manifest at the scales at which they enter an ecological system, but can also drive, via higher-order interactions^{49,50}, emergent effects at other scales. For example, pesticides can impair the foraging behaviour of bumblebees at the individual scale, which has emergent effects at higher scales by increasing the likelihood of whole-colony failure, impaired pollination services for multiple plant species and, ultimately, the loss of biodiversity⁵¹. While, in most cases, such emergent effects will occur at higher scales than that where the stressor entered the system, it is also possible for emergent effects to occur at lower scales. For example, density effects on populations have been linked to stress-hormone levels in surviving individuals with consequences for their behaviour and life history^{52–54}. Similarly, the arrival of a specialist invasive species, which enters at the community level by the addition of a new node in the species-interaction network, can, for example, have emergent effects on the population-scale targets of a competitor⁵⁵.

The impact of multiple stressors can be studied and modelled in line with the conceptualization outlined in Fig. 1. That is, embracing the idea that stressors 'enter' ecological systems at different scales, generate responses in targets and, through ecological complexity, generate responses at lower and higher scales. In other words, grouping stressors by what they do, not what they are, and model-

ling stressors in terms of the mechanistic targets where they act, not where they have their ultimate effect (for example, consider modelling climatic variation as a physiological-scale impact that has emergent effects at the population scale rather than as a term applied directly to a parameter in a population dynamics model⁵⁶). This allows bodies of ecological theory aligned with scales to cut across stressor identities and provide the platform on which to generate hypotheses about how and when stressor interactions arise.

Note that Fig. 1 represents just one example of stressor–target mapping and that other mappings that use different stressor classifications are certainly possible. In addition, the arrows shown in Fig. 1 represent all possible *a priori* connections between stressors and scales. Statistical analyses can provide valuable information about the relative importance/weighting of each of these arrows and thus a version of Fig. 1 incorporating weights or effect sizes for each arrow would likely have substantially fewer significant connections and reveal dominant scales at which stressors act. Figure 2 provides a framework for how such effect sizes can be estimated.

Below, we extend the interpretation of Fig. 1 with examples of how this framework can synthesize an understanding of multiple stressor effects by ecological scale and the targets at each of these scales.

Metabolism physiology and individual scales. These two scales are subpopulation scales that focus attention on traits of individuals. At these scales, ecological theory about trait-mediated effects frames the targets in which we detect how multiple stressors interact and the expectations about the patterns of these effects. These scales are vital components of a framework for assessing the magnitude and distribution of interactions among stressors because the strength of these sublethal effects is well known to be equal to, or stronger than, density-mediated effects in numerous communities^{31,57}.

The metabolism/physiology and individual scales also generate a substantial coalescence among stressors in disparate source categories (Fig. 1). For example, theory about thermal performance suggests that temperature change as a stressor makes its effects via altering metabolism^{29,58–62}. This focus on metabolism and physiology also encompasses, for example, how sublethal concentrations of contaminants can alter physiological processes (for example, neurological, reproductive and digestive modes of action, *sensu* ecotoxicological mode of action¹⁶). Responses to these types of stress at these two scales can generate emergent responses at the higher ecological scales of individual^{61–63}, population^{64–66} and community^{67–75}. For example, temperature or sublethal concentrations of pesticides have consequences on the population dynamics of species by affecting mortality and birth rates, life history traits (such as growth rates), reproductive performance, community structure and, potentially, diversity^{76–82}.

Population scale. Stressors causing mortality of individuals in specific populations are common, and assessments of stressor impacts on the targets of this scale (for example, density and biomass) receive significant attention. This scale also synthesizes among stressors from different sources, integrating classical theory about density-independent or density-dependent sources of mortality on populations^{83–85}. Descriptions of stressors at this scale are often more specific, with size-specific or trait-specific targets within a population. They are usually described as a loss function based on modifications of a functional response^{84,85} or reflecting specific equipment (for example, fishing gear^{86,87}). The sources of such stressor effects are most often aligned with the ‘overexploitation’ source (Fig. 1; for example, hunting or fishing⁸⁸), but are not limited to these sources. Lethal concentrations of pollutants¹⁷ and climate change can also directly generate mortality of individuals. As with the individual and metabolism/physiology scales, indirect consequences of effects at this scale emerge at higher ecological

scales^{49,89}. For example, abundance changes in producer or predator species are drivers of bottom-up and top-down effects, such as trophic cascades, which represent changes in relative abundance among multiple species at the community-scale. As exemplified in the fisheries-induced-evolution literature, the effects of harvesting stressors at the population scale can also emerge at the physiological and individual scale on life history traits^{86,87,90}.

Community scale. The community scale is defined by a species–interaction network, which encompasses species diversity and the distribution of interspecific interactions that define community structure. At this scale, targets include network properties such as connectance (network complexity), generalism and related descriptors of degree distributions, modularity, as well as other community-scale targets such as the body-size distribution.

There are at least three ways to frame the effects of multiple stressors at the community scale. The first frame centres on threats that directly target multiple populations within functional groups or trophic levels, but only indirectly the rest of the community. One example is agricultural nitrogen and phosphorus run-off, which can be classed as acting on entire functional groups or trophic levels that encompass multiple species such as primary producers (autotrophs), or ‘basal species’, and decomposers^{91,92} (although, notably, nutrient enrichment can be a dominant stressor at a range of scales⁹³). While the source of stress can vary, the targets of such effects are the community-scale distributions of carrying capacity, growth rates and reproductive traits across species comprising the trophic levels or functional groups (for example, physiology stress¹³), rather than the values for just one species population. Such impacts are defined by ecological theory about how enrichment, carrying capacity and productivity (bottom-up effects) generate impacts on biomass across trophic levels^{94,95}, which ultimately influence dynamics and community structure (defined by a species–interaction network)^{91,92,96,97}. For example, fertilizer run-off has contributed to enhanced primary production and a consequent depletion of oxygen in bottom waters, resulting in ‘dead zones’ in over 400 marine systems⁹⁸. Of course, similar patterns can happen via broad-spectrum chemical stressors that cause trophic cascades by reducing grazer abundance^{45,99,100}.

The second frame centres on threats that operate across functional groups and trophic levels, such as habitat loss and fragmentation caused by urbanization, deforestation, certain types of harvesting (for example, fishing) or bushfires. Again, the sources vary, but the defining feature of such threats is that populations of several species across functional groups and trophic levels are perturbed or removed from a community. While challenging, the body of ecological theory that helps us evaluate multiple threats centres on the decomposition of the contribution of mortality from multiple stressors on multiple species and the turnover components of beta-diversity^{40,101}.

The third frame is invasive species, which are defined by the arrival of a new species in the community. Invasive species therefore operate slightly differently to the majority of community-scale stressors by initially directly acting on species richness and network structure (a new node being added) with a multitude of indirect emergent effects at other scales, such as decreases in the fruit set of a competitor⁵⁵.

Ecosystem function scale. Our framework does not include direct effects of stressors on the ecosystem function scale. We narrowly define this scale as comprising targets, such as nutrient cycling and energy fluxes, that only respond to stress indirectly as a function of changes to targets at lower scales. For example, deforestation may change ecosystem-function-level nutrient cycling, but this is an emergent property of a reduction in the abundance, or change in relative abundance, of multiple tree species populations.

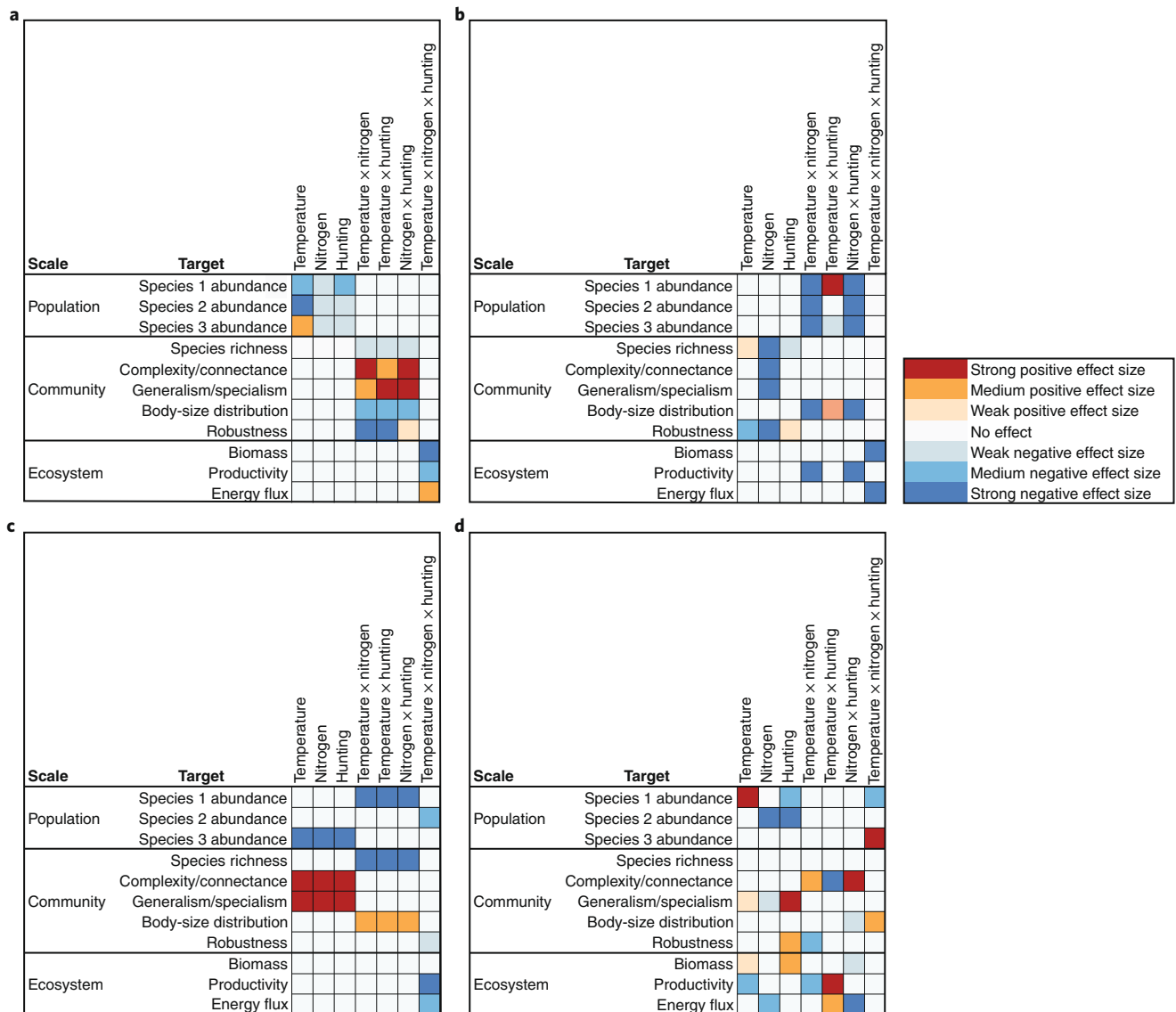


Fig. 2 | A framework for assessing the consistency and predictability of stressors. Rows represent targets at different scales of ecological organization. These are the variables that may be of interest when assessing the response of ecosystems to multiple, simultaneous stressors. Columns represent different stressors or their combinations. When a stressor is by itself (for example, temperature), this means that the stressor combines additively with other stressors. When a stressor occurs with another (for example, temperature $\times$ nitrogen), this means that the effects of one stressor depend on the other (interaction). Colours represent the magnitude and direction of the effect of a stressor, or combination of stressors, on each target. **a**, Stressors are more predictable at the population scale because they all combine additively, whereas prediction may be harder at the community and ecosystem scales due to complex two-way and three-way interactions. Consistency by scale is also visible: at a given scale, all stressors combine in the same way. For example, all stressors combine additively at the population scale, and all stressors are involved in two-way interactions at the community scale. **b**, Consistency by stressor, whereby a stressor has a similar effect on targets whether combining additively or interactively with other stressors. In this case, nitrogen consistently has a strong negative effect (all columns involving the nitrogen stressor have the same strongly negative effect size). **c**, Consistency by target, whereby targets are affected similarly by different stressors. For example, the abundance of species 3 is similarly affected by temperature, nitrogen and hunting. **d**, No pattern of consistency or predictability. Note that this figure is a caricature to demonstrate different forms of consistency and there is no intention to suggest the specific patterns shown are likely to occur in nature. For example, interactive effects can occur at all scales, not just at the community and ecosystem scales as shown in **a**. Moreover, we focus on three scales purely for visual simplicity; in a real analysis, the metabolism/physiology and individual scales are also important to include.

Evaluating the predictability and consistency of multiple stressor effects

Having outlined this classification of stressors, we now introduce how we can use this classification to understand the predictability and consistency of interactive effects.

Prediction is essential to assess the effects of different threats on ecosystems and to aid policymakers and practitioners in their

decision-making about ecosystem management^{102,103}. A key determinant of the predictability of stressor effects is whether stressors combine additively or whether they interact^{8,104}. If stressors combine additively, predicting their effects may be comparatively straightforward (although nontrivial in absolute terms), as it only requires a relationship between the intensity of individual stressors and their effect size on the target of interest. Conversely, if there are

Box 1 | Estimation of effect sizes

Calculating the effect sizes necessary to populate the grid framework in Fig. 2 can be accomplished with a variety of approaches, outlined below. The capacity for the Fig. 2 grid to deliver inference on predictability and consistency, however, requires attention to two details. First, the targets (row names) must be well defined and be distributed across multiple ecological scales. Second, the effect sizes must capture both magnitude and directionality.

One approach builds from the directional classification system of stressor interactions developed by Piggott et al.⁸ through to specific null model frameworks, such as the use of ecotoxicological null models of concentration addition and independent action^{19,125}. There are several examples of the use of the response ratios that define these null models to analyse empirical experiments¹³. For example, Galic et al.¹³ present an example of how these effect sizes can be used to make inference about the predictability and generality of multiple stressor effects. They used an individual-based model to explore the impacts of multiple ‘physiological’ stressors on individual traits with consequences estimated at biomass and ecosystem function scales. This framework typically requires an *a priori* experimental design that allows response ratios to be constructed using appropriate null models or control conditions. It is useful for both empirical and simulated *in silico* experiments.

An alternative to experimental approaches is to use effect sizes from standardized coefficients of linear models, generalized linear models and mixed-effects models. For example, Tabi et al.¹²⁶ combined this approach with the Piggott et al.⁸ framework to analyse the combined effect of temperature and productivity, aligning classical statistical definitions of interactions with estimations of effect sizes from coefficients. Meta-analyses are another approach, whereby studies can be synthesized to estimate interaction effect

sizes. For example, O’Brien et al.¹²⁷ use meta-analysis to estimate interactions between two stressors detected in targets classified by structure and function.

Multivariate auto-regressive state space (MARSS) models^{128,129} offer an additional toolbox for empirical time series data, empirical spatial data or data derived from stochastic simulations. MARSS models fit multiple regression equations simultaneously to each response variable, using the values of other variables (for example, species) and environmental covariates (the stressors) from previous time steps as predictors. A matrix of environmental covariate effect sizes (main and interactions) on response variables is recovered. These effect sizes are, by definition, associated with main effects of, and interactions among, stressors and are therefore suitable for use in the Fig. 2 grid concept. The MARSS models have an additional advantage in that they are good at partitioning uncertainty to understand its specific sources.

Didham et al.¹¹ and Geary et al.¹³⁰ have argued for expanding our understanding of the mechanisms of stressor interactions—beyond simply additive, synergistic or antagonistic—to also include different interaction pathways. The main two pathways are chain interactions, where one stressor alters the prevalence of another stressor directly, and modification interactions, where one stressor changes the prevalence and per capita effect of another stressor. Geary et al.¹³⁰ put forward additional statistical tools we can use to understand these pathways in the form of the ‘threat web’ approach, which uses a network approach to understand co-occurrence patterns among threats. The co-occurrence network can be analysed to gain quantitative insight into stressor–stressor interactions, and their causative or coincidental associations. In turn, this can frame our understanding of spatial and temporal threat co-occurrence¹³⁰.

two-, three- or higher-way interactions between stressors, their impacts become substantially more complicated to predict due to high levels of context dependence and whether the interactions can be defined on a linear scale (such as by a linear model). For example, if the effects of temperature on the abundance of a species depend on the levels of nitrogen and copper, forecasting is more difficult than if temperature acted independently (additively) of these other stressors. In the context of multiple, simultaneous stressors, we therefore define the predictability of a stressor in terms of how many other stressors it interacts with and the statistical order of the interactions. For example, if stressors (S_1) combine additively, this is not an interaction (or is a zero-order interaction; $S_1 + S_2$) and is the most predictable. If two stressors interact, this is a first-order interaction ($S_1 \times S_2$), which is less predictable. If three stressors interact, this is a second-order interaction ($S_1 \times S_2 \times S_3$), and so on. Predictability is therefore inversely related to the statistical order of the interactions in which stressors participate.

Consistency refers to patterns in how stressor impacts are distributed across ecological scales and among stressors. Below, we introduce three forms of consistency in detail. The first is “consistency by scale”, which refers to whether there are consistent patterns in how stressor impacts are distributed across ecological scales. For example, two-way interactions among multiple stressors may represent the strongest effects at the population scale but additive effects dominate at the community scale. The second form, “consistency by stressor”, relates to whether a stressor has a similar magnitude effect on different targets. For example, temperature having a negative effect of similar magnitude on all targets that it affects. The third form, “consistency by target”, is where a target is affected similarly by different stressors; for

example, temperature, habitat loss and pollution all having a similar magnitude of impact on species richness.

Evaluating predictability and consistency. Evaluating predictability and consistency requires a way to quantify the statistical order (whereby order 0 = additive, while order > 0 = interactions), magnitude, direction and distribution among scales of stressor impacts. Although the focus of this study is on anthropogenic stress, which are generally considered to have negative impacts, effect sizes allow for both positive and negative impacts of stress (Fig. 2). Importantly, because effect sizes act as a common, standardized currency, they allow comparisons to be made among scales, within or between studies, or between theoretical predictions and field observations. Here, we have separated the use of effect size from the estimation of effect size, and we focus on their use first (see Box 1 for a description of estimation).

Readers are referred to Fig. 2 for a set of caricatures of a three-stressor (temperature, nitrogen and hunting), three-ecological-scale (population, community and ecosystem) example to complement the following paragraphs. Note that we focus on three scales purely for visual simplicity; in a real analysis, the metabolism/physiology and individual scales are also important to include. We chose these three scales because targets at these scales (for example, abundance and richness) are the ones most often used in the context of stressor effects on ecosystems and conservation. It is also important to note that, as shown in Fig. 1 and discussed above, the targets at the ecosystem scale capture emergent responses rather than being directly affected by stressors. In Fig. 2, we use very broad stressors, such as ‘temperature’. In a real use case, it would be necessary to be more specific about the spatial and temporal extent

of stressors; for example, clearing a single path of habitat might be distinguished from ongoing habitat fragmentation, or an individual bushfire might be distinguished from a fire regime (discrete versus continuous stress, see ref. 48).

Predictability can be seen in Fig. 2 in cases where effect sizes for additive effects (stressors occurring by themselves in columns 1–3) are greater than (darker) those for interactive effects (columns 4–7). For example, the pattern in Fig. 2a suggests that responses at the population scale are more predictable than at community or ecosystem scales because effect sizes are clustered as additive stressor impacts at the population scale but as interactive stressor impacts at higher ecological scales. This extreme clustering in Fig. 2a also defines a strong pattern of consistency by scale, whereby all responses at the population scale are additive, all community responses are two-way interactions and all ecosystem responses are three-way interactions. The clustering in Fig. 2a also suggests that temperature might be classed as a ‘dominant driver’ at the population scale due to its strong effect sizes. Note that this figure is a caricature to demonstrate different forms of consistency and, while Fig. 2a shows interactive effects only at the community and ecosystem scales but not the population scale, interactive effects may occur at all scales in nature²².

In contrast, Fig. 2b shows consistency by stressor. This is where a column, in this case nitrogen, has a similar impact across scales and stressor combinations. Regardless of whether nitrogen is acting additively or interacting with other stressors, it consistently generates a strong negative effect size across targets and scales (in Fig. 2b, all columns involving nitrogen have a strong negative effect size). Figure 2c shows consistency by target, whereby impacts on a particular row (that is, a particular target) are similarly strong for all stressors or interactions. For example, the abundance of species 3 is particularly vulnerable to all three stressors combining additively, while species richness is particularly sensitive to two-way interactions between stressors, and ecosystem biomass is not affected by any of the stressors. Figure 2d, in contrast to the others, displays limited evidence of predictability or consistency as it lacks discernible structure in the distribution of effect sizes either vertically or horizontally.

We believe that thinking about the distribution, magnitude and direction of stressor impacts on targets in this way will enable important inferences to be made about the predictability and consistency of multiple stressor effects and interactions that are relevant to both basic science and applied management. For example, if a stressor is responsible for strong interactions at multiple scales (consistency by stressor; for example, nitrogen in Fig. 2b), this consistency helps prioritize and target the allocation of scarce resources to mitigate impacts by focusing on this stressor. Alternatively, if one target, such as the abundance of a particular species, is consistently affected by multiple stressors (consistency by target; for example species 1 or 3 in Fig. 2c), then conservation action may be directed at that species.

Absolute predictability and consistency. By definition, predictability and consistency may vary with the number and identity of stressors considered. This means that determining an ‘absolute’ value of the predictability or consistency of stressors would require a study that encompasses all main sources of ecosystem stress, such as the LPI sources in Fig. 1. This is particularly feasible in a modelling context. Alternatively, synthesizing across studies with different combinations of stressors can also lead to absolute conclusions about predictability and consistency. For example, if stressors A and B are found to interact across many studies, each of which considers a different set of stressors alongside A and B, then this could be evidence of an ‘absolute’ interaction between A and B, and therefore an ‘absolute’ lower level of predictability than if the stressors combined additively. Similarly, if a stressor, A, is shown to have a consistent negative impact on a particular target across different studies, each

evaluating different sets of stressors, this could also be evidence of ‘absolute’ consistency. If the different studies consider the same sets of stressors, then results are comparable and this can be useful to establish consensus across space and time.

Importantly, determining ‘absolute’ predictability or consistency may not always be an appropriate aim. This is because all stressors do not co-occur in all places at all times; therefore, in the context of managing a particular site for conservation, it may be more useful to understand predictability or consistency *in situ* by just considering the set of stressors occurring at a particular site. Moreover, it is valuable in its own right to understand the context dependence of predictability and consistency. For example, are particular stressors always hard to predict or only in the presence of other particular stressors? Or does a particular stressor consistently generate negative impacts on a target or does this only happen when a certain set of stressors are present? Again, answering questions like these can generate essential insights for conservation and resource management under environmental change.

Simulation models for studying multiple stressor effects

The framework shown in the first section (Fig. 1) outlines conceptually how a modelling approach for evaluating the impacts of multiple simultaneous stressors could work: stressors enter the system at a range of ecological scales, with emergent effects on targets at a range of scales. Here, we build on this framework and propose seven criteria (Fig. 3, central box) that simulation models must satisfy to be able to represent stressors and biology in this way. These criteria are motivated by a recent specification of several critical attributes for predictive models¹⁰⁵ and more general guidance about modelling ecological communities that can be useful for multiple stressor research¹⁰⁶. As a point of reference, we also outline in Fig. 3 the extent to which several current modelling tools meet our criteria.

Models that meet the seven criteria are the platform for generating robust, simulated data for improving knowledge about the impacts of multiple, interacting stressors (although they must also be linked to empirical data and experiments; Box 2). The seven criteria are specifically associated with modelling that generates times series of biomass or abundance for multiple individuals or species who are embedded in some representation of a community. This means that the simulated data can be analysed to produce effect sizes that populate the statistical and graphical framework presented in Fig. 2; thus, the simulated data are central to generating theory about predictability and consistency. Ultimately, the predictability and consistency framework detailed in Fig. 2 emerges from statistical analyses of simulated time series produced by a model (or empirical data). Models can be parameterized, for example, based on allometries and known environment response curves (such as temperature), and then used in numerical, fully factorial, *in silico* experiments about effects of multiple simultaneous stressors^{13,68,69,107}. While the framework in Fig. 2 can be used with either empirical or simulated data, the difficulty of large-scale empirical experimentation with multiple stressors (for example, combinatorial explosion of stressor combinations and difficulty in detecting biotic interactions) means that simulation models play a key role in improving understanding, generalizations and developing theory.

A common feature of all the models in Fig. 3 is that they meet criterion 1, which encompasses the ability of stressors to affect targets at different ecological scales (Fig. 3, ‘access’ row in the table). This is a particularly important feature because stressors have often been modelled generically, acting at scales that are different to how they may actually enter ecosystems. For example, environmental perturbations might be modelled by adding stochastic noise terms to population dynamics equations even if these perturbations are meant to represent stressors that primarily act at lower or higher scales. Such a decision may limit inference about how and why interactions arise because the biology through which two stressors

Bioenergetic food web model

Mechanistic model of biomass dynamics of species in food webs; links metabolism, growth and consumption with body size; temperature scaling is considered

Access	Stressors have access points at metabolism/physiology, individual, population and community scales
Biodiversity	Species, trophic, metabolic class and feeding group diversity represented
Interaction diversity	Interspecific predation, intraspecific competition/density dependence, indirect effects
Rewiring	NA
Scalable	Allometry allows a few equations to represent high-diversity communities
Predictions	Combined effects of stressors are emergent properties at a variety of ecological scales
Feedback	Feedback between body-size distribution, structure and dynamics

IBM/dynamic energy budget

Individual based, dynamic energy budget model; often focused on physiological mode of action

Access	Stressors have access points at metabolism/physiology, individual, population and community scales (limited)
Biodiversity	NA
Interaction diversity	Physiology and intraspecific density dependence
Rewiring	NA
Scalable	NA
Predictions	Combined effects of stressors are emergent properties at a variety of ecological scales
Feedback	NA

The seven criteria

(1) Access	Possess multiple access points or targets for stressors to generate their effects
(2) Biodiversity	Represent multiple species, size classes or metabolic classes, spanning multiple trophic or feeding groups
(3) Interaction diversity	Represent multiple types of interaction types, direct (e.g., predation/competition) and indirect (e.g., trophic cascades) effects
(4) Rewiring	Allow species losses or insertions to drive mechanistic change in the distribution of links among species as resource options change
(5) Scalable	Represent high diversity systems with a small number of equations, e.g., through allometry
(6) Predictions	The combined effects of stressors generate responses at the scales they enter and at higher or lower ecological scales
(7) Feedback	Allow changes in, and feedback between, traits, structure and dynamics

Trophic network model

Mechanistic model of tri-trophic biomass dynamics; links metabolism, growth and consumption to temperature and CO₂ scaling

Access	Stressors have access points at metabolism/physiology, individual, population and community scales
Biodiversity	Three trophic levels (limited)
Interaction diversity	Interspecific predation, intraspecific competition/density dependence, indirect effects
Rewiring	NA
Scalable	NA
Predictions	Combined effects of stressors are emergent properties at a variety of ecological scales
Feedback	NA

Size spectra

Mechanistic model of density per unit mass per unit volume for organisms of mass m at time t ; integrates metabolism, body size, consumption and growth; temperature scaling is considered

Access	Stressors have access points at metabolism/physiology, individual, population and community scales
Biodiversity	Size spectra maps on to species diversity and trophic groups; species-resolved versions exist
Interaction diversity	Interspecific and intraspecific predation, intraspecific competition/density dependence, indirect effects
Rewiring	NA
Scalable	Allometry allows a few equations to represent high-diversity communities
Predictions	Combined effects of stressors are emergent properties at a variety of ecological scales
Feedback	Feedback between body-size distribution, structure and dynamics

Fig. 3 | Criteria that models must satisfy to simulate the impacts of multiple stressors. We propose seven criteria (central blue box) that models must satisfy to generate informative, simulated data from which effect sizes can be estimated to quantify the impacts of multiple, interacting stressors. Models that meet these seven criteria provide a platform for developing a general theory for the predictability and consistency of multiple stressor effects. The criteria are motivated by a recent specification of several critical attributes for predictive models¹⁰⁵. We are currently unaware of any model that meets all seven criteria; here, we present four that have been used to evaluate multiple stressor effects on ecological communities and that possess several of the criteria. The bioenergetic food web model (top left; ^{116,123}) and the size spectra modelling (bottom right) frameworks^{118,124} are two that offer extensive opportunities because they meet many of the criteria and may potentially accommodate rewiring: the re-allocation of network links associated with loss or gain of species in the network (see the main text). The trophic network model¹⁰⁷ (bottom left) and the individual-based model¹³ (IBM; top right) examples are notable because of the detail specified on how various stressors might have an impact at different ecological scales. The following references motivated each criteria: (1) access¹¹⁶; (2) biodiversity^{116,124}; (3) interaction diversity^{116,124}; (4) rewiring^{108,109,112,111}; (5) scalable¹¹⁶; (6) predictions^{13,69,107}; and (7) feedback^{116,124}. NA, not applicable.

interact (for example, how higher-order interactions emerge) is not defined.

We note that no model in Fig. 3 currently incorporates adaptive foraging processes that allow species to rewire their connec-

tions in the network as the biomass of resources change or species go extinct or arrive, such as through invasion (criterion 4)^{108,109}. The central hypothesis is that species that can adjust their diets might persist and buffer ecological communities against stress. There are

Box 2 | The model–experiment interface

Model development (whereby models are defined as simulation models like those discussed in the section “Simulation models for studying multiple stressor effects”) must be linked with empirical data and experiments. While there is a rich history of empirical work linked to such efforts on multiple stressors (see “Introduction”), there remains a substantial set of biases along with major differences in semantics and design constraints that often limit the contribution of empirical work to these kinds of models^{7,14}. We suggest that recognizing at least three facets of the model–experiment interface can help reduce these biases, generate common semantics and maintain a strong relationship between models and data.

One facet of the interface between models and experiments focuses on experiments that produce data to parameterize models. Experiments help evaluate assumptions made by modelling and refine the identity and functional forms of mechanisms embedded in models. This is embodied, for example, in recent meta-analyses and syntheses of data on metabolism and functional response parameters, which are then directly used by simulation models^{29,131,132}.

A second facet of the interface is to align model predictions with experiments. Our grid approach in Fig. 2 requires effect sizes to populate the matrix elements. These effect sizes can be generated by analysing data produced by experiments and by simulation models. For experimental data to be compatible with our approach, we have four key recommendations. First, experiments must incorporate three or more stressors. Second, they must incorporate three or more scales of ecological organization. These two recommendations are an essential progression beyond the one-scale and two-stressors studies that currently dominate¹⁴. Third, studies must adopt a multifactorial experimental design that examines different stressor combinations. Fourth, the experiment

must be based around time series, as these are necessary to capture population dynamics that are then analysed to calculate the effect sizes required by the grid framework in Fig. 2. This also allows different stressor temporal trends to be incorporated, such as sudden versus ongoing changes in temperature⁴⁸.

Recent work suggests that experiments are increasing in design complexity and the number of stressors and scales evaluated^{133,134}. There are already a number of studies that adopt our recommendations and would be suitable for our framework. For example, Rillig et al.¹³³ experimentally manipulated soil communities, applying up to ten stressors. They found complex changes in soil properties, soil processes and microbial communities (that is, multiple scales of response) and decreasing predictability as the number of stressors increased (interactions were pervasive). Tabi et al.¹²⁶ experimentally manipulated temperature and productivity in microbial communities, whereby targets at multiple scales were monitored through time. The authors combined this experiment with an effect size analysis of the multiple stressor effects to estimate predictability and consistency across scales. Experiments like these are critical to match similar advances occurring in the mathematical modelling.

A third facet relates to ‘fitting models to data’, which can prove challenging, but beneficial, in cases where the complexity of the required experiments to estimate parameters is high. There are emerging methods that link large-scale empirical survey data with models to help parametrize them. These include ‘inverse methods’ or state-space models^{135,136} (often now Bayesian with priors to appropriately capture uncertainty). This is perhaps the least developed sphere of the model–experiment interface for the models in Fig. 3. However, it is also one with great potential as we move towards leveraging the insight from models to help manage natural resources and biodiversity in the future^{137–140}.

several rewiring models for food webs that could be incorporated into existing modelling frameworks. Historically, rewiring has been investigated in the context of secondary extinctions in food webs and explored by rules of thumb, inspired by concepts of foraging biology^{110,111}. More recently, optimal foraging theory has emerged as a formal principle for wiring and rewiring networks, specifically via the allometric diet breadth model^{112–114}. The allometric diet breadth model uses optimal foraging theory and allometries of foraging variables to predict food-web structure. The model predicts which resources a consumer should choose based on which food items maximize energy intake, subject to foraging constraints, such as encounter rates and handling times. The model performs well on empirical data, correctly predicting up to ~85% of links¹¹⁵, and has been used in topological modelling of primary and secondary extinctions¹¹².

The bioenergetic food web model and the size spectra model both meet all criteria except for incorporating rewiring (criterion 4), and thus offer extensive opportunities. The bioenergetic food-web model¹¹⁶ is an established, mechanistic model of biomass dynamics of species in food webs that integrates metabolism, body size, intra-specific density-dependence and interspecific interactions. In the model, parameters specifying metabolism, production of biomass (reproduction and population growth) and consumption are determined by body size via a set of allometries (each with a constant and exponent). These parameters determine rates of change in populations of a species—modelled as collections of biomass—via a set of ordinary differential equations solved numerically. Size spectrum models offer a size-structured alternative and are primarily used in the marine environment^{117,118}. A size spectrum is the relationship

between log individual size (body mass) and log abundance, which is roughly -1 . Size spectrum models therefore conceptualize food webs as a gradient of individuals, organized by size and other traits, whereby individuals grow by eating smaller individuals and die by being eaten by larger individuals^{117,119}. The core of the model is a set of equations (partial differential equations) that model the density per unit mass per unit volume for organisms of mass m at time t . Both these modelling platforms are temporally resolved and structured around some characterization of multi-species community structure. This mix of temporal resolution and structure is central to making inference about predictability and consistency and for developing theory about the effects of discrete versus continuous stress outlined by Jackson et al.⁴⁸ (such as the distinction between the gradual increase in mean temperature change and extreme temperature events).

Finally, we note that relying on a single model to generate theory and inference about predictability and consistency of multiple stressor effects can be risky; it may be beneficial to amalgamate inference from an ensemble of models to establish consensus. Many of the models discussed in Fig. 3 can share common state variables and could therefore be used as part of an ensemble modelling approach. Ensemble modelling helps deal with three main forms of uncertainty: process uncertainty, caused by the variation and stochasticity inherent to ecosystems; model uncertainty, related to the components of a particular model; and future/scenario uncertainty, related to predictions being made, such as from uncertainty about future conditions¹⁰⁶. Ensemble modelling has recently gained traction in ecology and has already been used to evaluate the impacts of multiple stressors on particular targets, including explicitly building

relationships between stressors into the models^{120,121}. To capture a final type of uncertainty, known as parameter uncertainty (uncertainty related to the parameter values used in a model¹⁰⁶), ensemble modelling can be combined with global sensitivity analysis. Global sensitivity analyses can help understand the sensitivity of results and conclusions to changes in model parameters by, for example, running models with parameter values obtained from bootstrap resampling from parameter distributions^{106,122}. Thus, ensemble modelling could potentially have a large part to play in multiple stressor research in the future.

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Competing interests

The authors declare no competing interests.

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